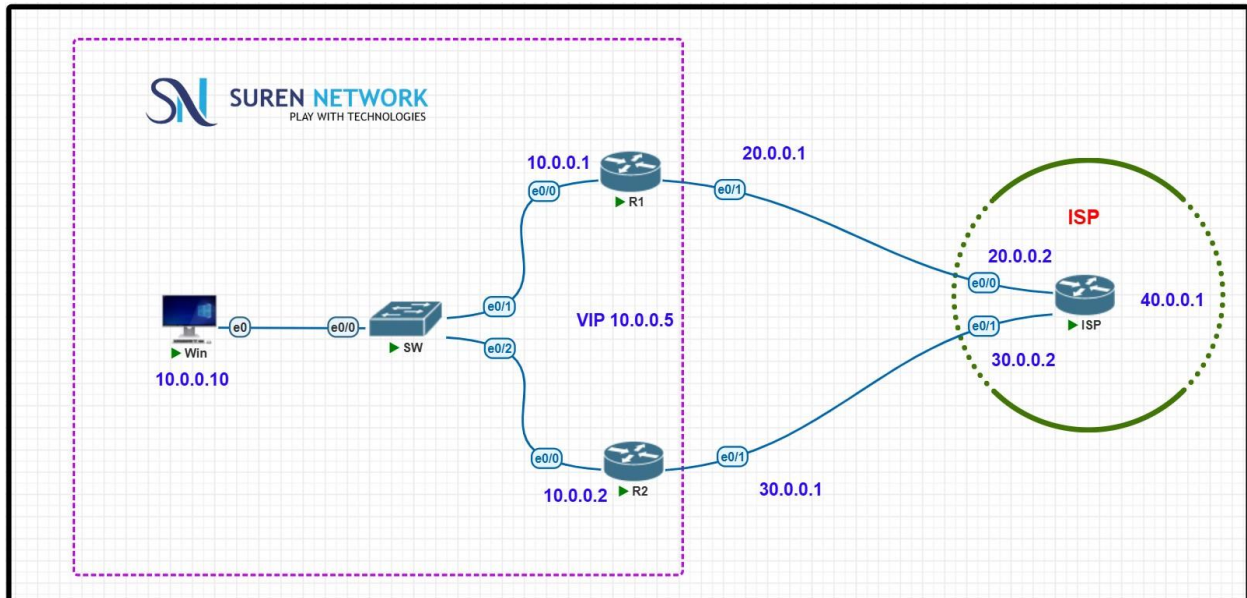


HSRP



Enable HSRP on lan interface in router-1 and router 2

```

R1
R1(config)#int e0/0
R1(config-if)#standby 1 ip 10.0.0.5
R1(config-if)#ex
R1(config)#
R1(config)#do sh standby
Ethernet0/0 - Group 1
  State is Listen
  Virtual IP address is 10.0.0.5
  Active virtual MAC address is 0000.0c07.ac01
  Local virtual MAC address is 0000.0c07.ac01 (v1 default)
  Hello time 3 sec, hold time 10 sec
  Preemption disabled
  Active router is 10.0.0.2, priority 100 (expires in 8.688 sec)
  Standby router is unknown
  Priority 100 (default 100)
  Group name is "hsrp-Et0/0-1" (default)
R1(config)#
*Sep 22 08:03:37.933: %HSRP-5-STATECHANGE: Ethernet0/0 Grp 1 state Speak -> Standby
R1(config)#do sh standby
Ethernet0/0 - Group 1
  State is Standby
  1 state change, last state change 00:00:13
  Virtual IP address is 10.0.0.5
  Active virtual MAC address is 0000.0c07.ac01
  Local virtual MAC address is 0000.0c07.ac01 (v1 default)
  Hello time 3 sec, hold time 10 sec
  Next hello sent in 1.168 secs
  Preemption disabled
  Active router is 10.0.0.2, priority 100 (expires in 9.840 sec)
  Standby router is local
  Priority 100 (default 100)
  Group name is "hsrp-Et0/0-1" (default)
R1(config)#

R2
do sh standby
Ethernet0/0 - Group 1
  State is Active
  2 state changes, last state change 00:00:32
  Virtual IP address is 10.0.0.5
  Active virtual MAC address is 0000.0c07.ac01
  Local virtual MAC address is 0000.0c07.ac01 (v1 default)
  Hello time 3 sec, hold time 10 sec
  Next hello sent in 2.224 secs
  Preemption disabled
  Active router is local
  Standby router is unknown
  Priority 100 (default 100)
  Group name is "hsrp-Et0/0-1" (default)
R2(config)#
  
```

Assign ip address for lan pc with default gateway as VIP

```
QEMU (Win)
C:\WINDOWS\system32\cmd.exe
C:\Documents and Settings\user>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address . . . . . : 10.0.0.10
    Subnet Mask . . . . . : 255.0.0.0
    Default Gateway . . . . . : 10.0.0.5

C:\Documents and Settings\user>
C:\Documents and Settings\user>
```

Ping 20.0.0.2 from lan pc & check router failover

```
graph LR
    PC[10.0.0.10] --- SW[SW]
    SW --- R1[10.0.0.1]
    SW --- R2[10.0.0.2]
    R1 --- VIP[VIP 10.0.0.5]
    R2 --- VIP
```

```
QEMU (Win)
C:\WINDOWS\system32\cmd.exe
C:\Documents and Settings\user>ping 20.0.0.2

Pinging 20.0.0.2 with 32 bytes of data:

Reply from 20.0.0.2: bytes=32 time=2ms TTL=254
Reply from 20.0.0.2: bytes=32 time=1ms TTL=254
Reply from 20.0.0.2: bytes=32 time=1ms TTL=254
Reply from 20.0.0.2: bytes=32 time=1ms TTL=254

Ping statistics for 20.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Documents and Settings\user>tracert 20.0.0.2

Tracing route to 20.0.0.2 over a maximum of 30 hops

  0  1 ms  <1 ms  <1 ms  10.0.0.2
  1  1 ms  <1 ms  <1 ms  20.0.0.2

Trace complete.

C:\Documents and Settings\user>
```

```
graph LR
    PC[10.0.0.10] --- SW[SW]
    SW --- R1[10.0.0.1]
    SW --- R2[10.0.0.2]
    R1 --- VIP[VIP 10.0.0.5]
    R2 --- VIP
```

```
QEMU (Win)
C:\WINDOWS\system32\cmd.exe
C:\Documents and Settings\user>ping 20.0.0.2

Pinging 20.0.0.2 with 32 bytes of data:

Reply from 20.0.0.2: bytes=32 time=2ms TTL=254
Reply from 20.0.0.2: bytes=32 time=1ms TTL=254
Reply from 20.0.0.2: bytes=32 time=2ms TTL=254
Reply from 20.0.0.2: bytes=32 time=1ms TTL=254

Ping statistics for 20.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Documents and Settings\user>tracert 20.0.0.2

Tracing route to 20.0.0.2 over a maximum of 30 hops

  0  1 ms  <1 ms  <1 ms  10.0.0.1
  1  1 ms  <1 ms  <1 ms  20.0.0.2

Trace complete.
```